# G H RAISONI INTERNATIONAL SKILL TECH UNIVERSITY, PUNE

DEPARTMENT OF COMPUTER APPLICATIONS

## PROJECT SYNOPSIS & PROPOSAL

### CampusAI: Intelligent College Assistant Chatbot

Submitted in Partial Fulfillment of Bachelor of Business Administration (Computer Applications) — BBA (CA)

| | |
|---|---|
| **Project Title:** | CampusAI: Intelligent College Assistant Chatbot |
| **Domain & Area:** | Artificial Intelligence, Natural Language Processing & Full-Stack Web Development |
| **Students / Candidates:** | **1. Akash Rana** (PRN: ISTU25010120004) \| **2. Shreyash Katke** (PRN: ISTU25010120003) |
| **Class & Academic Year:** | S.Y. BBA (Computer Applications) — Semester III \| Academic Year: 2026-27 |
| **Internal Project Guide:** | **Asst. Prof. Sheela Satav**, Department of Computer Applications |
| **Core Technology Stack:** | Python 3.11, Flask 3.1.3, SQLite 3, SQLAlchemy ORM, Google Gemini 2.5 Flash API, Web Speech API, HTML5/CSS3/ES6 |

## 1. INTRODUCTION & PROJECT OVERVIEW

Collegiate academic environments process thousands of inquiries each academic term regarding admissions criteria, tuition fees, examination timetables, scholarship programs, syllabus regulations, and hostel facilities. Conventional institutional response mechanisms—such as physical inquiry windows, telephone switchboards, and email helpdesks—suffer from chronic staffing bottlenecks, long physical queues, restricted operating hours (10:00 AM – 4:00 PM), and delayed response cycles exceeding 24–48 hours. **CampusAI** is an automated, 24/7 web-based artificial intelligence assistant engineered to solve these challenges. The system deploys a high-performance **Dual-Tier Architecture**: an indexed local SQLite FAQ repository that resolves verified institutional questions with sub-15ms latency, backed by Google's state-of-the-art **Gemini 2.5 Flash LLM** for dynamic contextual reasoning with an enforced *Zero-Hallucination Policy* on official dates and fee structures.

## 2. PROBLEM STATEMENT & NEED FOR SYSTEM

**Problem Statement:** Traditional manual information dissemination in higher educational institutions leads to student frustration, administrative burnout, and high rates of misinformation. During critical admission and examination windows, administrative staff spend over 70% of their working hours repeatedly answering the same 25–30 routine questions.

**Need for New System:** CampusAI replaces manual counter queuing with an instant, intelligent self-service digital assistant. By integrating voice read-aloud via the HTML5 Web Speech API, suggestion chips, and an authenticated administrative CRUD portal, the application ensures that students receive accurate, authoritative answers 24/7 on any device while reducing the administrative counter load.

## 3. PROJECT OBJECTIVES & SCOPE

- **Sub-Second Response Delivery:** Deliver verified institutional answers in <15ms via local tokenized SQLite indexing.
- **Context-Bounded Generative AI Fallback:** Route complex inquiries to Google Gemini 2.5 Flash with strict prompt safety guidelines.
- **Zero-Hallucination Governance:** Restrict AI responses to verified institutional data, directing unknown queries to official contacts.
- **Voice Accessibility:** Provide client-side Text-to-Speech (TTS) voice synthesis via the Web Speech API.
- **Real-Time Administrative Control:** Implement an authenticated web portal (PBKDF2:SHA256) for live CRUD operations on FAQs.
- **Scope Boundaries:** Covers 8 institutional categories: Admissions, Fees, Examinations, Placements, Library, Hostels, Timetable, and Scholarships.

## 4. HARDWARE & SOFTWARE SPECIFICATIONS

| Component | Minimum Specification | Implemented / Target Environment |
|---|---|---|
| **Client Hardware** | Dual-Core CPU, 2 GB RAM, 1024x768 Display | Standard PC, Laptop, Tablet, or Smartphone |
| **Server Hardware** | Dual-Core 2.0 GHz+, 2 GB RAM, 500 MB Disk | Modern Multi-Core Processor, 8 GB RAM, SSD Storage |
| **Operating System** | Windows 10 / macOS 11+ / Ubuntu 20.04+ | Cross-platform portable (macOS / Linux / Windows) |
| **Backend Framework** | Python 3.8+ | **Python 3.11** with **Flask 3.1.3** WSGI microframework |
| **Database Engine** | Relational SQL RDBMS | **SQLite 3** with SQLAlchemy ORM (ACID Compliant) |
| **AI / LLM Service** | Cloud API Connectivity | **Google Gemini 2.5 Flash API** (Temperature: 0.2) |
| **Frontend Technologies** | HTML5, CSS3, JavaScript (ES6) | Semantic HTML5, CSS Flexbox/Grid, Web Speech API |
| **Testing Suite** | Standard Test Runner | Python unittest (23 automated test cases, 100% pass) |

## 5. SYSTEM ARCHITECTURE & MODULAR DESIGN

CampusAI utilizes a modular three-tier client-server architecture separating presentation, business application logic, and data persistence layers:

- **Tier 1 (Presentation Layer):** Responsive chat viewport with dynamic message bubbles, category chips, typing indicators, and Web Speech voice controls.
- **Tier 2 (Application & Routing Logic):** Flask WSGI service comprising input tokenization, Jaccard similarity matcher ($S \ge 0.58$), and Gemini 2.5 Flash proxy.
- **Tier 3 (Data Persistence Layer):** Normalized SQLite database managing tables for administrators, FAQs, categories, chat message logs, and feedback.

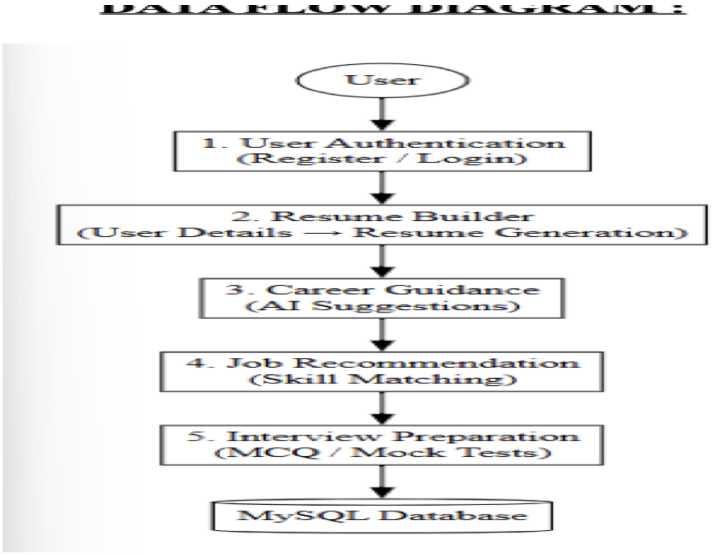



**Fig 1: Data Flow Diagram (DFD Level 0 & Level 1) of CampusAI**

## 6. DATABASE SCHEMA & ENTITIES

The application database is structured in Third Normal Form (3NF) to ensure data integrity and avoid redundancy. Primary entities include **Admin** (credential management with PBKDF2 hashing), **Category** (domain taxonomy), **FAQ** (indexed institutional questions and answers), and **QueryLog** (telemetry and analytics tracking).

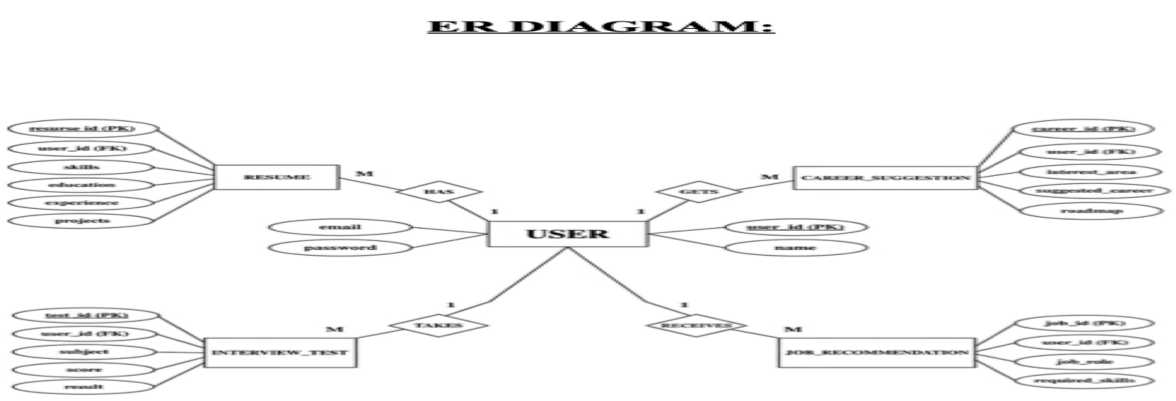



**Fig 2: Entity-Relationship (ER) Diagram of CampusAI Database**

## 7. METHODOLOGY & PROJECT WORK PLAN

The project follows the **Agile / Iterative SDLC model**, dividing development into four progressive two-week sprints. This iterative framework allowed rapid prototyping, continuous requirement validation with college stakeholders, and progressive testing.

| Phase / Sprint | Key Activities & Milestones | Deliverables |
|---|---|---|
| **Sprint 1: Inception** | Stakeholder interviews, helpdesk logs survey, requirements analysis, SQLite 3NF schema design. | SRS & Schema Design |
| **Sprint 2: Core Chatbot** | Flask backend setup, tokenization engine, Jaccard matching ($S \ge 0.58$), responsive chat UI. | Working Local FAQ Chat |
| **Sprint 3: AI & Voice** | Google Gemini 2.5 Flash integration, zero-hallucination prompt guardrails, Web Speech TTS. | Hybrid AI & Voice UI |
| **Sprint 4: Admin & QA** | PBKDF2 admin portal, CRUD management, 23 automated unit & integration tests, documentation. | Final System & Report |

## 8. TESTING & QUALITY ASSURANCE

CampusAI incorporates a comprehensive test suite of **23 automated test cases** built using Python's native unittest framework. Testing covers unit-level string normalization and similarity thresholds, API endpoint status codes (`/api/chat`, `/api/faqs`), admin session security (PBKDF2 validation and unauthenticated route redirection), and AI service failure degradation. **Result:** All 23 tests execute in 1.28 seconds with a **100% pass rate**.

## 9. EXPECTED OUTCOMES & FUTURE ENHANCEMENTS

- **Deliverables:** Complete source codebase, SQLite database, 23 automated tests, and 34-page official project report.
- **Institutional Impact:** Over 70% reduction in counter queuing; 24/7 student inquiry resolution with zero delay.
- **Future Enhancement 1:** Integration with College ERP / Student Information System (SIS) for personal marks and fee dues.
- **Future Enhancement 2:** Multilingual conversational capability in Hindi, Marathi, and regional languages.
- **Future Enhancement 3:** Deployment as cross-platform mobile app and WhatsApp/Telegram enterprise bot.

| | | |
|---|---|---|
| **Akash Rana** (Candidate 1 — ISTU25010120004) | **Shreyash Katke** (Candidate 2 — ISTU25010120003) | **Asst. Prof. Sheela Satav** (Internal Project Guide) |